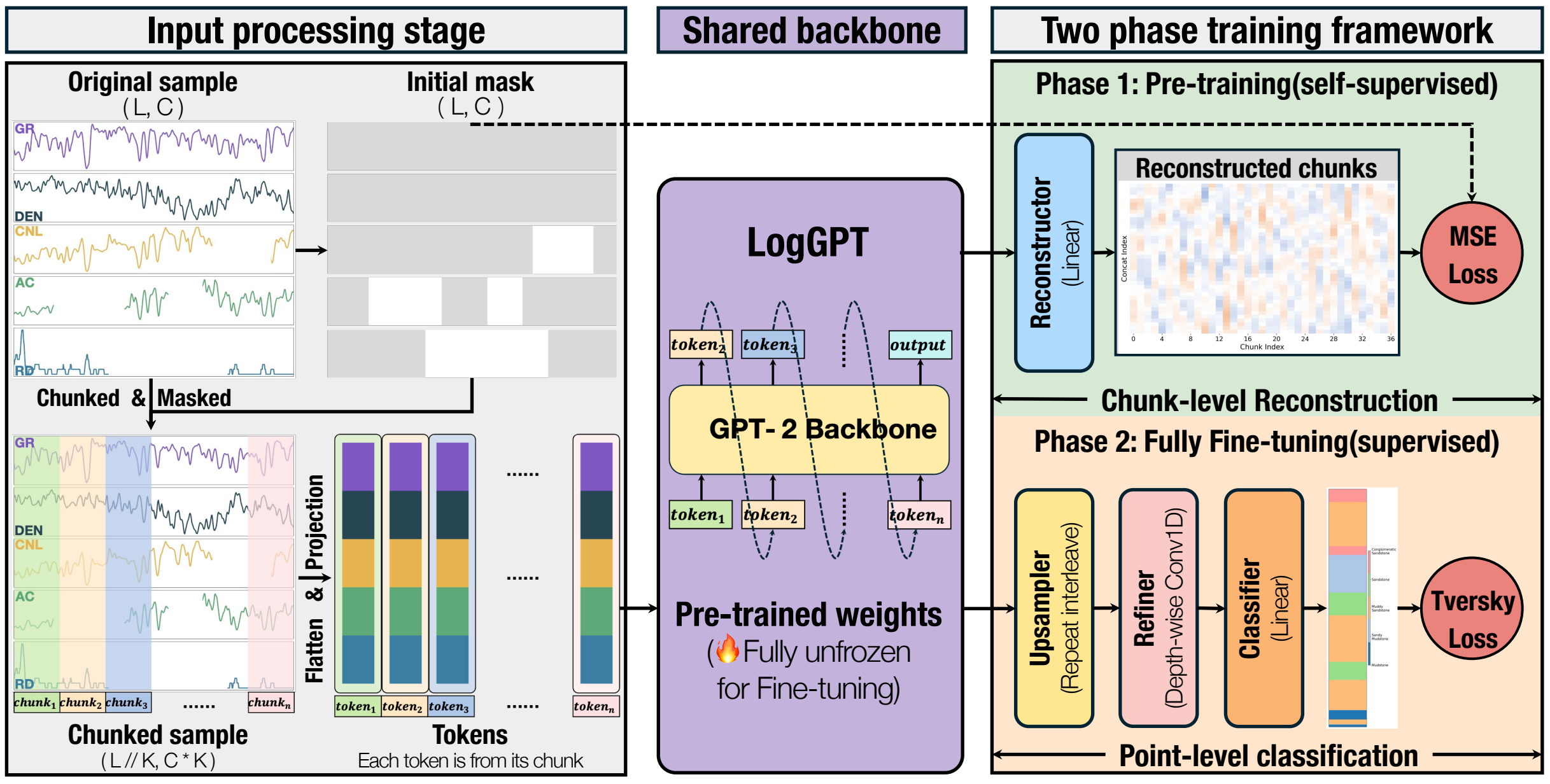
Input processing stage
Shared backbone
Two phase training framework
Original sample
(L, C)
Initial mask
(L, C)
GR
DEN
CNL
AC
RD
Chunked & Masked
Flatten & Projection
chunk1 chunk2 chunk3 ..... chunkn
Chunked sample
(L // K, C * K)
token1 token2 token3 ..... tokenn
Tokens
Each token is from its chunk
LogGPT
token2 token3 output
GPT- 2 Backbone
token1 token2 tokenn
Pre-trained weights
(🔥Fully unfrozen
for Fine-tuning)
Phase 1: Pre-training(self-supervised)
Reconstructor
(Linear)
Reconstructed chunks
Concat Index
Chunk Index
MSE
Loss
Chunk-level Reconstruction
Phase 2: Fully Fine-tuning(supervised)
Upsampler
(Repeat interleave)
Refiner
(Depth-wise Conv1D)
Classifier
(Linear)
Tversky
Loss
Point-level classification